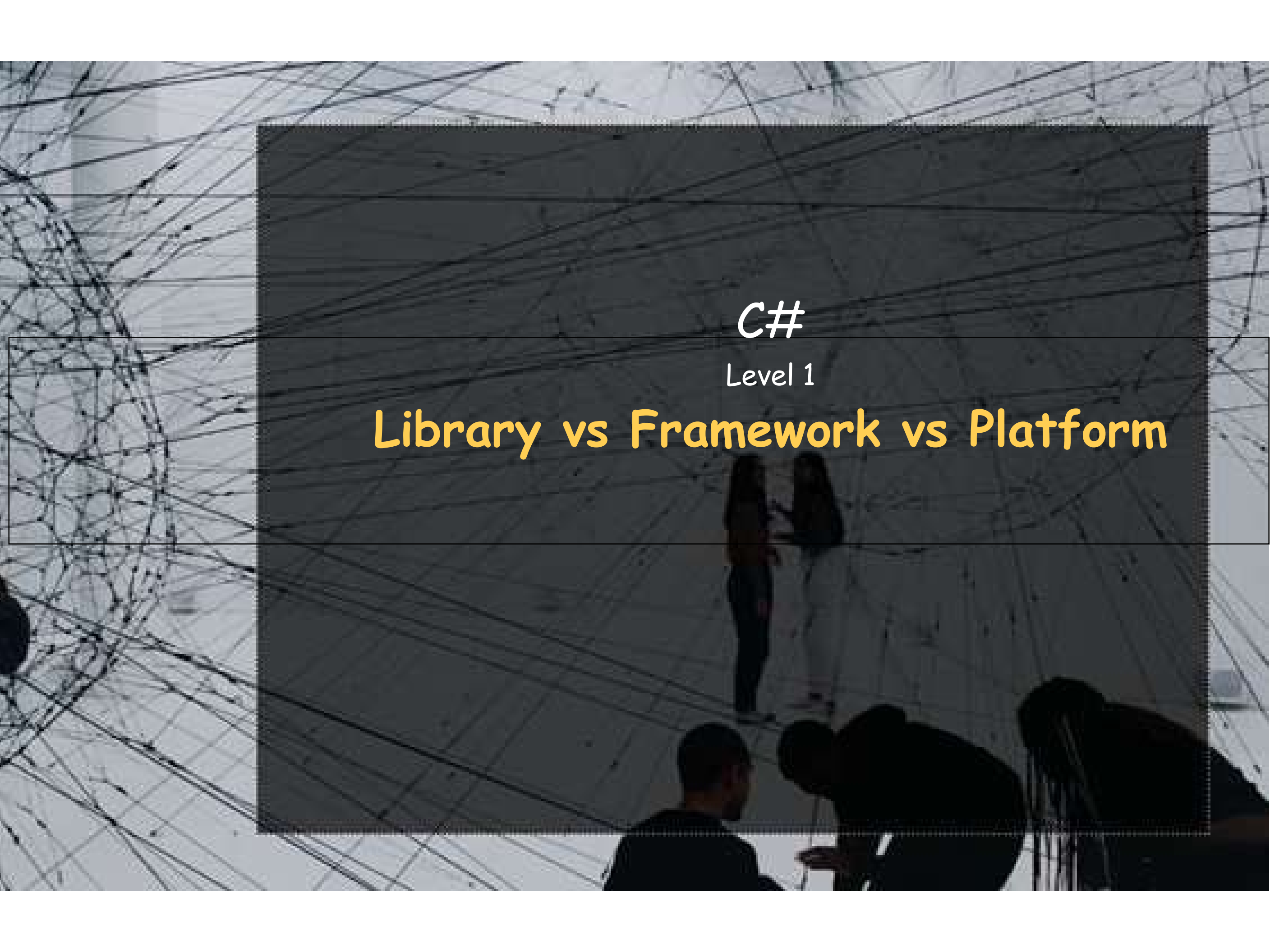




C#

Level 1

Library vs Framework vs
Platform



C#

Level 1

Library vs Framework vs Platform

Definition

Library

-  A Library is a collection of reusable code that your program can call when needed.

-  The developer controls the application flow.

Framework

-  A Framework provides a structure for building an application.

-  The framework usually controls the overall application flow.

Platform

-  A Platform is a broader environment that provides the runtime, tools, system capabilities, and services needed to develop and run software.

Explanation

Library

-  Your code calls the library.

-  Examples include C++ STL, NumPy, and jQuery.

Your Code



Calls the Library



Library performs a task

➡ You call the library.

Framework

-  The framework provides the application structure.

-  Your code is written inside that structure.

-  Examples include Django, Spring, and ASP.NET Core.

Framework



Controls the application flow



Your Code

➡ The framework calls your code.

🔍 Definition

■ Platform

▨ The platform provides the environment where software is developed and runs.

▨ Examples include Windows, Linux, Android, and the .NET platform.

Platform



Provides the environment



Framework / Libraries



Application

➡ The platform provides the environment where software runs.

Procedures

1. Using a Library

-  Include or reference the library.
-  Call the required function or API.
-  Continue controlling your program.

2. Using a Framework

-  Create a project using the framework.
-  Follow its structure.
-  Implement your application logic.
-  Let the framework manage the application flow.

3. Using a Platform

-  Choose the target platform.
-  Use its available tools and services.
-  Build the application.
-  Run the application on that platform.

💡 Example

Suppose you are building a web application:

Platform



.NET / Windows / Linux



Framework



ASP.NET Core



Libraries



JSON / Database / Logging Libraries



Your Application

- ▨ Platform → provides the environment
- ▨ Framework → provides the application structure
- ▨ Library → provides reusable functionality

Note

- ◆ The easiest way to remember the difference is:

Library → You call it

Framework → It calls your code

Platform → Your software runs on it

- ◆ Think of building a house:

▨ Library → Tools you choose to use

▨ Framework → The structure and rules for building

▨ Platform → The land and infrastructure where the house exists

⊘ Warning

- ◆ A library is not necessarily a framework.
- ◆ A framework can contain or use many libraries.
- ◆ A platform is broader than both and may provide operating-system features, runtimes, development tools, and other services.



Comparison

▨ Library

- ✓ Provides reusable functionality
- ✓ You call the library
- ✓ Gives you more control over application flow
- ✓ Example: C++ STL

▤ Framework

- ✓ Provides application structure
- ✓ Framework controls the application flow
- ✓ You write code according to its structure
- ✓ Example: Django

▥ Platform

- ✓ Provides the environment for software
- ✓ Includes runtime and system capabilities
- ✓ Applications run on the platform
- ✓ Example: Windows / Linux

✦✦ Summary

- ✓ Library provides reusable functionality that your code calls
- ✓ Framework provides a structure and controls the application flow
- ✓ Platform provides the environment where software is developed and runs
- ✓ Library = You call it
- ✓ Framework = It calls your code
- ✓ Platform = Provides the environment

Thank you 😊